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Quiz 1

(Standard deduction of -1 for wrong identification of scalar and vector applies everywhere.)

Q1

$$\vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$$= \hat{x}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \hat{y}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)$$

$$+ \hat{z}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)$$

[0.5 marks if anyone has done correctly till this point]

$$\therefore \vec{\nabla}\cdot\left(\vec{\nabla} \times \vec{A}\right)$$

$$= \frac{\partial}{\partial x}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \frac{\partial}{\partial y}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)$$

$$+ \frac{\partial}{\partial z}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)$$

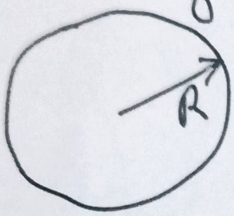
[1.5 marks]

$$= \cancel{\frac{\partial^2 A_z}{\partial x \partial y}} - \cancel{\frac{\partial^2 A_y}{\partial x \partial z}} + \cancel{\frac{\partial^2 A_x}{\partial y \partial z}} - \cancel{\frac{\partial^2 A_z}{\partial y \partial x}}$$

$$+ \cancel{\frac{\partial^2 A_y}{\partial z\, \partial x}} - \cancel{\frac{\partial^2 A_x}{\partial z\, \partial y}}$$

$$= 0$$ $\Leftarrow$ [4 marks]

Q2.



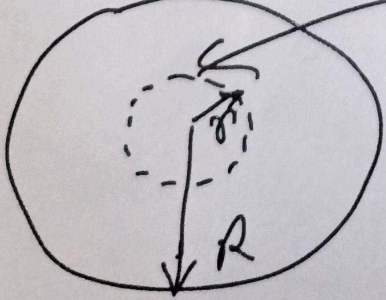
$\sigma = \frac{Q}{4\pi R^2}$

Charge will be distributed uniformly on the surface of the metal sphere. So, no charge can stay inside

This statement is important (−0.5 if this isn't written)

For $r < R$:

Gaussian surface

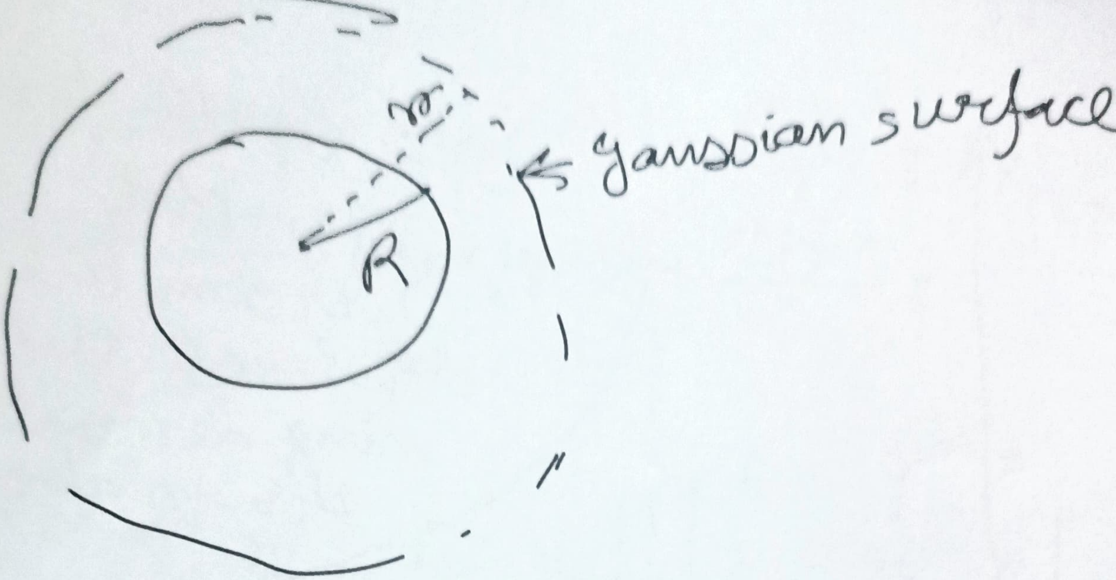


$$\oint \vec{E} \cdot \vec{dS} = \frac{Q_{enc}}{\varepsilon_0}$$

$$= 0$$

$\therefore E = 0.$ $\Leftarrow$ 2 marks

For $r > R$:



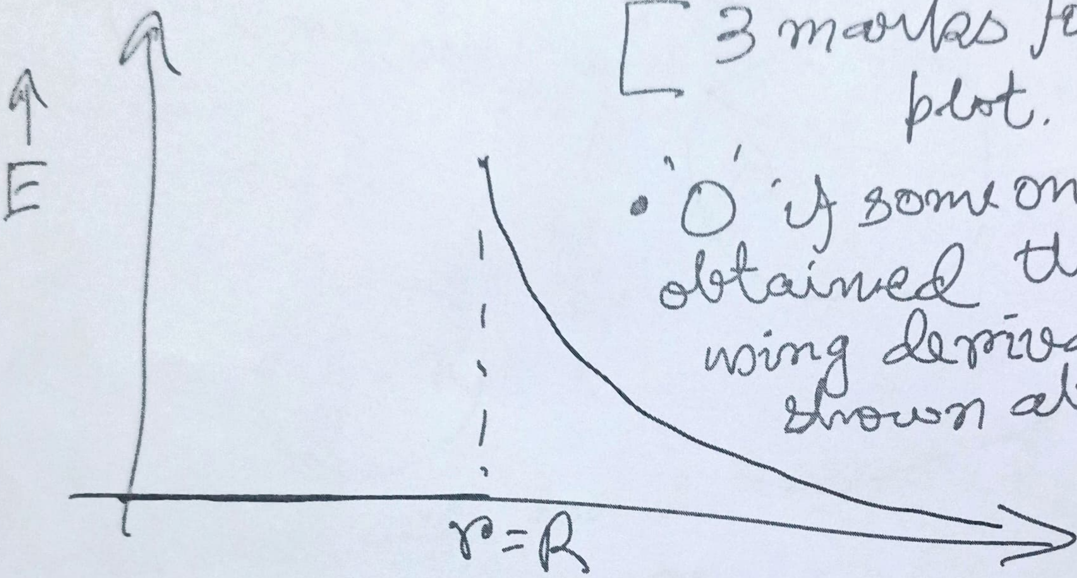
$$\oint \vec{E} \cdot \vec{ds} = \frac{Q}{\varepsilon_0}$$

$$\Rightarrow \quad E\, 4\pi r^2 = \frac{Q}{\varepsilon_0}$$

$$\Rightarrow \quad E = \frac{Q}{4\pi \varepsilon_0 r^2}$$

$$\vec{E} = \frac{Q\hat{r}}{4\pi \varepsilon_0 r^2}$$

Either of these 2 marks (But $\vec{E} = \frac{Q}{4\pi\varepsilon_0 r^2}$ is wrong and would be treated as absence of vector sign (standard rule of −1 applicable)



[3 marks for this plot.

- '0' if someone hasn't obtained the fields using derivation shown above)

- −1 for missing axis labels along any of x, y axis or both.

- −1 if the point $r = R$ is missing

Q3)

$$\bar{\nabla} F = \left(\frac{\partial}{\partial x}\hat{x} + \frac{\partial}{\partial y}\hat{y} + \frac{\partial}{\partial z}\hat{z}\right)F$$

$$= \left(\frac{\partial F}{\partial x}\hat{x} + \frac{\partial F}{\partial y}\hat{y} + \frac{\partial}{\partial z}\hat{z}\right)$$

No part marking

$$= \log_e y\left(\frac{e^x}{x} + \log_e x \; e^x\right)\hat{x} + \frac{e^x \log_e x}{y}\hat{y} + 0$$

$$= \frac{\log_e y \; e^x}{x}\left(1 + x\log_e x\right)\hat{x} + \frac{e^x \log_e x}{y}\hat{y}$$

$$\left(e^x \ln x \ln y + \frac{e^x \ln y}{x}\right)\hat{x} + \frac{e^x \ln x}{y}\hat{y}$$

Q5)

$$\bar{\nabla} \times \bar{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A_x & A_y & A_z \end{vmatrix} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ yz & 4xy & y \end{vmatrix}$$

No part marking

$$= \left[\frac{\partial}{\partial y}(y) - \frac{\partial}{\partial z}(4xy)\right]\hat{x} - \left[\frac{\partial}{\partial x}(y) - \frac{\partial}{\partial z}(yz)\right]\hat{y}$$

$$+ \left[\frac{\partial}{\partial x}(4xy) - \frac{\partial}{\partial y}(yz)\right]\hat{z}$$

$$= [1-0]\hat{x} - [0-y]\hat{y} + [4y-z]\hat{z}$$

$$= \hat{x} - y\hat{y} + (4y-z)\hat{z}$$

Q6) $\phi = \frac{4}{x} + \frac{3}{y^2} - \frac{2}{z}$

(i) $\vec{E} = -\vec{\nabla}\phi$

$= -\left[\frac{\partial}{\partial x}\left(\frac{4}{x}\right)\hat{x} + \frac{\partial}{\partial y}\left(\frac{3}{y^2}\right)\hat{y} - \frac{\partial}{\partial z}\left(\frac{2}{z}\right)\hat{z}\right]$

$= -\left[-\frac{4}{x^2}\hat{x} - \frac{6}{y^3}\hat{y} + \frac{2}{z^2}\hat{z}\right]$

No part marking $= \frac{4}{x^2}\hat{x} + \frac{6}{y^3}\hat{y} - \frac{2}{z^2}\hat{z}$

2

(ii) $\vec{\nabla}\cdot\vec{E} = \frac{\rho}{\epsilon_0}$

$\Rightarrow \frac{\partial}{\partial x}\left(\frac{4}{x^2}\right) + \frac{\partial}{\partial y}\left(\frac{6}{y^3}\right) - \frac{\partial}{\partial z}\left(\frac{2}{z^2}\right) = \frac{\rho}{\epsilon_0}$

$\Rightarrow \frac{\rho}{\epsilon_0} = -\frac{8}{x^3} - \frac{18}{y^4} + \frac{4}{z^3}$

$\Rightarrow \rho = \epsilon_0\left(-\frac{8}{x^3} - \frac{18}{y^4} + \frac{4}{z^3}\right)$

2

No part marking.

Q. 7

a) (i) conservative
(ii) irrotational

b) (ii) electrostatic field inside a charge-free dielectric sphere

c) (i) $E_{t1} = E_{t2}$

Surface must be close.

(iii) $\varepsilon_1 E_{n1} = \varepsilon_2 E_{n2}$

d) (iii) $\int_V (\vec{\nabla} \cdot \vec{E})\, dv = \oint_S \vec{E} \cdot \vec{dS}$